

Report

SBA 930 - Evaluating LLMs for Financial Forecasting and Risk Analysis



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Overview

This report evaluates ChatGPT and Gemini for financial forecasting and risk analysis using historical economic, stock-market, and commodity-index data. The evaluation examines forecast usefulness, relevance, reliability, risk identification, mitigation strategies, and the effect of structured prompt optimization.

The analysis is educational and uses the project datasets and saved model responses. It is not investment advice.

Two financial scenarios were used:

1. Stock-market and economic scenario: Historical GDP growth, inflation, and market-index information were supplied to ChatGPT and Gemini for analysis of potential stock-market effects.
2. Commodity-price scenario: Historical commodity-index information was supplied to both models for analysis of possible future trends.

Both LLMs received comparable prompts for each scenario. Their responses were saved in the results folder and reviewed for accuracy, relevance, reliability, transparency about uncertainty, and whether claims were supported by the available data.

Economic forecasts were also compared with historical observations through the project backtesting exercise. The comparison files were used to identify whether model outputs aligned with the supplied historical data and whether conclusions remained appropriately qualified.

Stock-Market Forecasting

ChatGPT and Gemini both identified that changes in GDP growth, inflation, and market-index values can be relevant context for stock-market forecasting. Each model was able to summarize historical patterns and discuss possible implications for future market conditions.

ChatGPT provided a structured response that linked economic indicators and market-index changes, but its conclusions required careful review because historical relationships do not establish causation. Gemini also produced useful trend-oriented observations, although forecast statements similarly required validation against the supplied data.

Neither model should be treated as producing a verified market forecast from the available dataset alone. The data is limited in time span and does not include company-level financials, interest rates, policy details, trading volume, or other variables needed for a complete forecast.

Commodity Forecasting

Both models analyzed the historical commodity-index pattern. The commodity index moved from 100 in 2019 to 92 in 2020, increased to 118 in 2021 and 135 in 2022, and then declined to 127 in 2023.

The responses were useful for describing the observed pattern, including a decline, a subsequent increase, and a later decrease. However, neither model could reliably establish the causes of these movements from the provided data. Forecast conclusions were therefore more reliable when stated as conditional scenarios rather than definite predictions.

Model Comparison

Evaluation Area	ChatGPT	Gemini
Relevance	Produced relevant discussion of economic, market, and commodity trends	Produced relevant discussion of historical trends and possible future conditions
Strength	Clear structure and detailed explanations	Useful alternate interpretation of the same data
Limitation	Can sound confident even when data does not support causal claims	Also requires validation because trend descriptions do not prove forecast accuracy
Reliability	Improved when the prompt requires evidence and limitations	Improved when the prompt requires evidence and limitations
Best use	Initial analysis, scenario framing, and structured risk discussion	Cross-checking interpretations and identifying differences between model outputs

Overall, using two models improved the evaluation because it allowed outputs to be compared rather than relying on a single response. Agreement between models can identify common observations, but it does not prove that a forecast is correct.

Economic Trend Forecasting and Backtesting

The project used GDP growth and inflation data to assess economic-trend forecasts. GDP growth was 2.3% in 2019, -3.4% in 2020, 5.9% in 2021, 1.9% in 2022, and 2.5% in 2023. Inflation was 1.8% in 2019, 1.2% in 2020, 4.7% in 2021, 8.0% in 2022, and 4.1% in 2023.

The historical data shows substantial variation across years. For example, GDP growth moved from -3.4% in 2020 to 5.9% in 2021, while inflation reached 8.0% in 2022 before declining to 4.1% in 2023.

The backtesting comparison showed that LLM forecasts must be judged against actual historical observations rather than accepted at face value. Forecast reliability is limited by the small dataset, changing economic conditions, and the absence of important explanatory variables. Therefore, LLM-generated forecasts should be treated as hypotheses for review rather than as confirmed estimates.

Risk Identification and Mitigation

Market-Volatility Risk

The market index changed from 2700 in 2019 to 2300 in 2020, rose to 3100 in 2021, declined to 2900 in 2022, and increased to 3400 in 2023. These year-to-year movements provide direct evidence of changing market conditions.

The data does not contain a formal volatility measure, trading volume, market breadth, individual-security returns, or interest rates. Therefore, the causes and full extent of market volatility are not supported by the available data.

Mitigation actions include diversifying exposures and establishing exposure limits that are reviewed when large market-index movements occur.

Credit Risk

GDP growth was negative at -3.4% in 2020. This can be treated as an observation of economic stress, but the dataset does not include defaults, delinquencies, credit ratings, credit spreads, debt levels, probability of default, or loss-given-default measures.

Credit-risk conclusions are therefore limited. A possible hypothesis is that worsening economic conditions could be associated with credit deterioration, but the available dataset cannot prove that this occurred.

Mitigation actions include conducting stress tests based on observed GDP changes and setting credit-exposure limits using actual borrower financial information, default history, and credit-quality data.

Liquidity Risk

The dataset shows large market and commodity-index changes, including a market-index decline from 2700 in 2019 to 2300 in 2020 and a commodity-index increase from 92 in 2020 to 135 in 2022.

However, it does not provide trading volume, bid-ask spreads, market depth, transaction costs, cash balances, funding availability, or redemption activity. Actual liquidity conditions are not supported by the available data.

Mitigation actions include maintaining sufficient liquid resources and establishing liquidity limits based on actual cash requirements and asset-liquidity characteristics.

Optimization and Enhancement

A controlled prompt-optimization test was applied to the ChatGPT financial-risk analysis. The optimized prompt required a consistent structure for each risk category: directly supported evidence, unsupported conclusions, assumptions, potential impacts, early-warning indicators, mitigation actions, and a confidence rating.

The prompt also required the model to use only the provided dataset, avoid causal claims, label unproven impacts as hypotheses, and explicitly state “Not supported by the available data” when information was missing.

The optimized output improved transparency and traceability. It cited specific data values, separated evidence from assumptions, listed missing credit and liquidity measures, and added a human-review checklist. This reduced the likelihood that unsupported claims would be presented as facts.

However, optimization does not create new data. Direct credit-risk and liquidity-risk conclusions remain limited without default rates, credit spreads, trading volume, bid-ask spreads, cash balances, and funding measures.

LLM Limitations

LLMs can generate useful summaries and structured analyses, but they have important limitations in financial applications:

- They may produce plausible statements that are not supported by the supplied data.
- They can imply causal relationships when the data shows only correlation or coincident timing.
- Outputs can vary depending on the wording and detail of the prompt.
- They cannot replace quantitative validation, backtesting, or domain-expert judgment.

- Limited historical datasets may not capture changing market conditions or rare events.
- LLM outputs require verification before being used in a real financial decision.

Ethical Considerations

Responsible use of LLMs in financial analysis requires transparency, accountability, and human oversight.

- Analysts should disclose that the output was generated with LLM assistance.
- Claims should be tied to provided evidence, and unsupported claims should be identified clearly.
- LLM output should not be presented as personalized investment advice.
- Human reviewers should validate assumptions, data quality, forecasts, and mitigation recommendations.
- Sensitive financial information should be protected and handled according to organizational privacy and security requirements.
- Model outputs should be documented so that decision-makers can understand the evidence, limitations, and reasoning used.

Recommendations

- Use structured prompts that require evidence, assumptions, limitations, confidence ratings, and human review.
- Use more complete data for future analyses, including credit, liquidity, interest-rate, trading-volume, and company-level financial measures.
- Apply quantitative methods such as backtesting, error metrics, scenario analysis, and stress testing before relying on forecasts.
- Compare outputs from more than one model to identify inconsistent interpretations.
- Require subject-matter-expert review before any LLM-supported financial recommendation is used in practice.
- Continue documenting prompts, model outputs, assumptions, and validation results.

Steps

1. Selected two financial-data scenarios: a stock-market/economic-indicator scenario and a commodity-index scenario.
2. Selected two LLMs for comparison: ChatGPT and Gemini.
3. Created comparable prompts for both models to analyze:
 - The potential impact of economic conditions on stock-market trends
 - Historical commodity-index data to identify possible future trends
 - GDP growth and inflation data for economic-trend forecasting
 - Market-volatility, credit, and liquidity risks
4. Saved each model's responses in the project results folder for review and comparison.
5. Compared ChatGPT and Gemini outputs for accuracy, relevance, reliability, clarity, limitations, and support from the supplied data.
6. Backtested the economic-trend forecasts by comparing model forecasts with the provided historical GDP-growth and inflation values.
7. Identified market-volatility, credit, and liquidity risks. Developed mitigation actions and documented data limitations for each risk category.
8. Applied prompt optimization to the ChatGPT risk analysis by requiring structured evidence, explicit assumptions, missing-data guardrails, hypothesis labels, confidence ratings, and human-review requirements.
9. Compared the baseline and optimized ChatGPT risk outputs to assess whether the optimized prompt improved transparency, consistency, and reliability.
10. Documented LLM limitations, ethical considerations, recommendations, and supporting project artifacts in the final report.

Conclusion

ChatGPT and Gemini were useful for summarizing historical financial data, organizing possible scenarios, and identifying areas requiring further risk analysis. Their outputs were most reliable when prompts required direct evidence, explicit limitations, and clear separation between observations and assumptions.

The optimized ChatGPT prompt improved the quality of the risk analysis by increasing structure, transparency, and awareness of missing data. Nevertheless, LLMs should support—not replace—quantitative validation and qualified human judgment in financial forecasting and risk management.

Appendix

- `results/chatgpt\_stock\_market.txt`
- `results/gemini\_stock\_market.txt`
- `results/chatgpt\_commodity.txt`
- `results/gemini\_commodity.txt`
- `results/forecast\_comparison.md`
- `results/chatgpt\_economic\_backtest.txt`
- `results/gemini\_economic\_backtest.txt`
- `results/economic\_backtest\_comparison.md`
- `results/chatgpt\_risk\_analysis.txt`
- `results/gemini\_risk\_analysis.txt`
- `results/risk\_comparison.md`
- `results/chatgpt\_optimized\_risk\_analysis.txt`
- `results/optimization\_impact.md`